DLS Buddy — Quickstart Guide

Version 0.9.0

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1. Introduction

DLS Buddy is a general-purpose Python program for analyzing **dynamic (DLS)** and **static (SLS)** light scattering data. It is not tied to any specific instrument, solvent, or polymer system. All physical parameters — viscosity, refractive index, temperature, $\mathrm{d}n/\mathrm{d}c$, wavelength, and scattering angle — are always supplied by the user. Nothing is assumed or hard-coded.

The graphical interface is organized as a sidebar workspace (your loaded data, auto-grouped into samples) plus tabbed modules:

- **Data** — confirm and edit the physical parameters of each measurement.
- **DLS** — correlation-function fitting (cumulants, exponentials, KWW), CONTIN/NNLS distributions, and multi-angle / concentration-series analysis.
- **SLS** — Zimm, Berry, and Debye analysis for Mw, Rg, and A_2, with a calibration panel.
- **Cross-Sample** — quantities that link samples or modules: the shape parameter $\rho = R\text{g}/R\text{h}$, and log–log scaling plots.
- **Utilities** — data-quality diagnostics (intensity traces, I·sin θ) and a synthetic correlogram generator.
- **Settings** — global defaults and appearance.

A core design goal is to support mixed DLS+SLS sessions, linking Rh (from DLS) and Rg (from SLS) measured on the same sample to compute $\rho = R\text{g}/R\text{h}$. The internal data model supports this linkage directly.

Launching the program

Double-click the launcher for your operating system:

- **Windows** — `Launch DLS Buddy (Windows).bat`
- **macOS** — `Launch DLS Buddy (MacOS).command` (on first run macOS Gatekeeper may refuse an unsigned script — right-click it → **Open** → confirm once, and it opens normally thereafter)

Or run it from the repository root with:

```
python -m gui.main
```

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2. The Workflow at a Glance

Most analyses follow the same short path. Here is the whole flow in one place; the rest of the guide fills in the detail.

Launch → Load → Confirm parameters + Update → open the DLS or SLS tab → pick a method and Run → read the results → Export.

1. **Launch** the program (`python -m gui.main`, or the double-click launcher for your OS).
2. **Load** your data — **Load DLS correlogram...** or **Load SLS intensities...**. Files are auto-detected and auto-grouped into **samples** in the left sidebar (§3).
3. **Confirm the parameters** on the **Data** tab — temperature, viscosity, refractive index, wavelength, scattering angle, and (for SLS) dn/dc and concentration. Identity and optics are entered **once per sample**; press **Enter** to jump to the next field. Click **Update** to commit — analysis runs on the committed values (§6).
4. **Open the analysis tab** for what you need:
   - **DLS** (size, Rh): tick the measurement(s) in the sub-tab's list and choose a method — **Correlogram** (cumulant / exponential) or **Distribution** (CONTIN / NNLS / Lognormal) — then **Run**. The multi-angle Γ **vs q^2** and concentration-series **D vs c** tabs tick the points to include in the fit (§10.1).

- **SLS** (Mw, Rg, A_2): select the sample, set the **calibration**, choose **Zimm / Berry / Debye**, then **Run** (§10.2).

5. **Read the results** — each method fills a results **table** plus a plot; flags mark *apparent* vs *thermodynamic* and *uncalibrated* results so they aren't over-read.
6. **Export** with **Export CSV...** (Origin-compatible) (§11).
7. *(Optional)* Combine a sample's DLS Rh and SLS Rg on the **Cross-Sample** tab for $\rho = R\text{g}/R\text{h}$ and the scaling plots (§10.3).

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3. Loading Data

The program accepts data from multiple sources through a **two-layer parser architecture**. Instrument-specific parsers translate native file formats into a common internal structure; all analysis operates on that common structure. Adding a new instrument format never requires changes to analysis or export code.

Every file load follows the same workflow:

1. Click **Load DLS correlogram...** (or **Load SLS intensities...**) and pick the file(s) — you can select **several at once**, and each is auto-detected independently (any that can't be read are reported at the end, the rest still load). You do **not** choose an instrument — the program **auto-detects** the format, trying each supported instrument parser until one recognises the file.
2. The parser reads the file and pre-fills any parameters it can find.
3. The **Data** tab shows all fields. Pre-filled values can be changed; blank fields must be completed before analysis.
4. **Update** commits the parameters; analysis then runs on the committed values.

   Note. Pre-filled values from a file are suggestions, not locked inputs. You can always override what the parser found — for example to correct an instrument software value or to apply a calibration you determined independently. If a file is in none of the supported formats, the program says so rather than guessing.

3.1 Supported File Formats

Loading is **instrument-agnostic**: the load buttons first auto-detect among the instrument formats below (each sniffs its own format). If none match, the program falls back to a **plain two-column** file — for a generic DLS file it asks for the delay-time unit and data form (g_2-1 / g_2 / g_1); a generic SLS file (angle, intensity) needs no prompt. Only if that also fails does it report an unrecognised format.

Format	Parser	Produces
Brookhaven Particle Explorer DLS `.CSV`	`BrookhavenDLSParser`	One DLS measurement per file
Brookhaven Particle Explorer count rate `.CSV`	`BrookhavenTraceParser`	One intensity trace per file
Brookhaven Particle Explorer SLS `.CSV`	`BrookhavenSLSParser`	One measurement per concentration (solvent reference included)

Format	Parser	Produces
ALV correlator `.ASC` (e.g. ALV-7012 CGS-12F)	`ALVCorrelatorParser` / `ALVTraceParser`	**All angles** in the file as separate DLS measurements (one sample) + a count-rate trace per angle
Malvern Zetasizer clipboard correlation text	`ZetasizerClipboardParser`	**All records** in the file as separate DLS measurements (no metadata — all parameters entered at confirmation)
Malvern Zetasizer structured *export* `.txt`/`.csv`	`ZetasizerExportParser`	**One DLS measurement per row**; pre-fills refractive index, temperature, viscosity, and the sample/material/dispersant names (see §3.2)
Plain-text two-column DLS (delay, $g_2-1/g_2/g_1$)	`GenericDLSParser`	One DLS measurement per file (fallback; prompts for delay unit + data form)
Plain-text two-column SLS `(angle, intensity)`	`GenericSLSParser`	One concentration per file (fallback; loads directly)
Plain-text two-column trace `(time, count rate)`	`GenericTraceParser`	One intensity trace per file (Utilities ▶ Traces; prompts for units)

3.2 Malvern Zetasizer export template

The Zetasizer software's **export** is configurable, so the program reads its columns by **name** (not by position) and accepts any column order. To produce a file the export parser recognises, configure the exporter as follows:

- **Settings:** include the header row; use commas as the separator.
- **Parameters (any order):** Sample Name, Material Name, Dispersant Name, Dispersant RI, Temperature, Viscosity, Correlation Delay Times, Correlation Data.

The two correlation blocks (Delay Times + Data) are required; each named parameter is optional and pre-filled when present. The program maps the columns as:

Export column	Pre-fills	Conversion
Sample Name	sample label	—
Material Name	polymer/material name	—
Dispersant Name	solvent name	normalised to the solvent vocabulary
Dispersant RI	solvent refractive index	—
Temperature	temperature	°C → K
Viscosity	viscosity	cP → Pa·s
Correlation Delay Times	lag-time axis	µs → s
Correlation Data	correlogram (g_2-1)	stored as-is

Scattering **angle**, **wavelength**, and **concentration** are not in the export and are entered at the confirmation step, as with any DLS file.

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4. Brookhaven Particle Explorer File Formats

Brookhaven Particle Explorer exports DLS and SLS data as comma-separated `.CSV` files. The formats are described for reference; the parsers handle all reading automatically.

4.1 DLS Correlation Function (`.CSV`)

- Row 1: sample label string (e.g. `PnVP (40k) in Water - 6/1/2026 4:46:59 PM`).
- Row 2: column headers (`t(µs)`, `C(t)`, `Fitted (CONTIN)`).
- Rows 3+: delay time (µs), $g_2(\tau){-}1$, and the instrument CONTIN fit.
- Trailing padding rows with $t = 0$ and $C(t) = -1$ fill unused correlator channels; they are stripped automatically.

 Important. The DLS file contains no instrument metadata. Angle, wavelength, refractive index, temperature, viscosity, and sample name must all be supplied manually in the Data tab. **Concentration is optional for DLS** — cumulant, exponential, KWW, and CONTIN/NNLS size analysis (and Γ vs q^2) never use it; it is needed only for the multi-concentration diffusion extrapolation.

 Note. The program ignores the Brookhaven CONTIN fit column. All regularization is performed independently by the program's own pipeline.

The correlogram and count-rate history are separate files and are associated manually. (Sample labels in the two files sometimes differ slightly in spacing — a known Brookhaven quirk — which is why automatic pairing is not attempted.)

4.2 Count Rate History (`.CSV`)

- Row 1: sample label string.
- Row 2: column headers (`Time (seconds)`, `Count Rate (kcps)`).
- Rows 3+: elapsed timestamp (`HH:MM:SS.sssssss`) and count rate in kcps.

Timestamps are elapsed time from the start of the run, not wall-clock time. Count rates are converted to counts per second (cps) on load.

4.3 SLS Intensity Export (`.CSV`)

One file contains all concentrations and angles, as a flat list of key–value pairs: a metadata header, an angle table, a concentration table, and an intensity grid (`Intensity - Concentration N - Angle M, value`). The parser extracts all available metadata. Only two fields are never present and must be entered manually:

- Sample name.
- Temperature (the SLS export has no temperature field).

 Rayleigh ratio. The Brookhaven software embeds its own toluene Rayleigh ratio. It is read and recorded for traceability but **never used** in analysis — the program always computes the temperature- and geometry-corrected value from the toluene Rayleigh literature. See the Advanced Guide §7.5.

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5. Generic (Plain-Text) File Formats

The generic parsers accept data from any source — home-built correlators, simulation output, collaborator data, spreadsheet exports — as long as it is a clean two-column numerical table. Because these files carry no metadata, **all** physical parameters are supplied in the Data tab.

5.1 Generic DLS Format

File contract (the parser rejects non-compliant files):

- Exactly two columns of numbers; no header, comment, or blank lines.
- Comma- or tab-separated (auto-detected).
- Column 1: delay times (unit specified at confirmation).
- Column 2: correlation values (interpretation specified at confirmation).
- At least two data rows.

Delay-time units: `s`, `ms`, `us` (µs), `ns` — converted to seconds on load.

Column-2 interpretation (data form). The program converts every input to $g_2(\tau)-1$ (baseline-subtracted intensity autocorrelation) before storage:

Data form	Column 2 contains	Extra parameter
`g2m1`	$g_2(\tau)-1$, already baseline-subtracted	none
`g2`	$g_2(\tau)$, baseline not yet subtracted	baseline $B > 0$
`g1`	field autocorrelation $g_1(\tau)$	coherence factor $\beta \in (0, 1]$

For the `g2` form, the baseline is removed; for the `g1` form, the Siegert relation converts the field autocorrelation to the measured quantity (β is the instrument coherence factor, an optical property, typically 0.01–1.0 for standard goniometers, determined independently). See the Advanced Guide for the equations.

> **Tip.** Unsure which form you have? Check the range of column 2. $g_2(\tau)-1$ starts near the intercept β (often 0.3–1.0) and decays to 0; unsubtracted $g_2(\tau)$ starts near $1+\beta$ and decays to 1; $g_1(\tau)$ starts near 1 and decays to 0.

5.2 Generic SLS Format

Each file is one concentration at multiple angles (one file per concentration).

- Exactly two columns; no header/comment/blank lines; comma- or tab-separated.
- Column 1: scattering angles in degrees, each strictly in (0, 180).
- Column 2: raw intensity values.
- At least two data rows.

> **Column order.** Angle must be column 1, intensity column 2. If reversed, the parser detects column-1 values outside (0, 180)° and raises an error suggesting the columns may be swapped.

For a Zimm/Berry/Debye dataset, load one file per concentration (including the solvent reference at $c = 0$) and set each concentration in the Data tab. The SLS module assembles the dataset from the loaded measurements.

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6. Physical Parameters and Units

All quantities are **stored in canonical internal units**; you enter and read them in whatever unit is convenient. In the **Data** tab each parameter row has a **Unit** column: a dropdown for quantities with a choice of units (the value re-displays when you switch units — the stored physical value is unchanged), or a fixed label for quantities with a single natural unit. Defaults are chosen on a human scale.

Parameter	Internal unit	Input units (default first)
Concentration	g/mL	**mg/mL**, g/L, µg/mL, g/mL
Temperature	kelvin (K)	**°C**, K, °F
Viscosity	Pa·s	**mPa·s**, cP, Pa·s
Delay-time window (DLS tab)	seconds (s)	**µs**, ns, ms, s
Wavelength	nanometres (nm)	nm
Scattering angle	degrees	deg
dn/dc	mL/g	mL/g
Count rate (trace)	counts/second (cps)	**kcps**, cps, Mcps (chosen on load)

> **SLS intensities.** The calibrant intensity and dark count rate are entered in **the same units as your SLS intensity file** (whatever the instrument exported). The program does not convert them, because only the *ratio* of calibrant to sample intensities enters the Rayleigh ratio — a fixed-unit conversion there would introduce a scale error. The fields are labelled "(file units)" accordingly.

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7. Solvent Names

Solvent identity is part of sample identity: the same polymer at the same concentration and temperature in two solvents is two distinct datasets (enabling, e.g., PVP-in-water vs. PVP-in-reline comparisons). Names are normalized against a controlled vocabulary, so `Water`, `water`, and `H2O` are equivalent. Unrecognized names are accepted (used lowercased) with a one-time warning; to add one permanently, edit `_SOLVENT_ALIASES` in `core/data_models.py`.

> **Note.** Sample-name matching is case-sensitive (`PVP` ≠ `pvp`); solvent-name matching is case-insensitive and alias-aware. (The sample name is stored internally as the `polymer_name` field — samples need not be polymers.)

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8. How the Program Organizes Data

Every measurement carries a sample key — (polymer, solvent, concentration, temperature) — used to associate related datasets, for example to pair a DLS and an SLS measurement of the same sample when computing $\rho = R\text{g}/R\text{h}$. For grouping into a **sample**, the identity is (polymer, solvent, rounded temperature); concentration and angle are axes *within* a sample.

8.1 Molecular-weight fractions

A scaling study measures several **molecular-weight fractions** of one polymer (same chemistry, solvent, and temperature), each with its own concentration series. Because molecular weight is *not* part of the sample identity, all such fractions group into one sample. To keep them separate, give each measurement an optional free-text **Mw fraction** label (e.g. `250k`, `1M`) in the Data tab; editing it offers to apply the label to every concentration loaded from the same file. The label is *not* required — leave it blank for single-fraction samples or systems with no meaningful Mw (e.g. nanoparticles).

Within a sample, the fraction label **partitions the SLS analysis**: the SLS tab shows a *Fraction* selector and runs an independent Zimm/Berry/Debye fit for the selected fraction, while the single c = 0 solvent reference is **shared** across all fractions. Each fraction therefore yields its own $M\text{w}$, $R\text{g}$, $A2$ (and, from DLS, $R\text{h}$). In the Cross-Sample tab each fraction contributes its own point to the scaling plots ($R\text{g}$–$M\text{w}$, $A2$–$M\text{w}$) and its own row in the ρ table, so a molecular-weight series produces a proper scaling curve rather than a single merged point. (The per-fraction Cross-Sample views build on the per-fraction SLS analysis described above.)

Intensity traces (count-rate histories) are independent objects: they can be loaded and analyzed without a correlogram, or linked to a DLS measurement after the fact — supporting batch diagnostics over a concentration series.

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9. Data Quality and Utilities

The Utilities module assesses data quality (especially for intensity traces) and generates synthetic test data. These help identify slow-mode fluctuations, drift, and dust before committing to full correlation-function or Zimm analysis.

9.1 Intensity-Trace Diagnostics

- **Summary statistics & baseline.** Mean, standard deviation, coefficient of variation, min, max. The baseline (true scattering level) is estimated robustly so upward dust spikes do not bias it — by a lower percentile (default 25th) or a sigma-clipped mean.
- **Outlier flagging.** Points outside the shot-noise envelope are flagged. For a Poisson (shot-noise-limited) process the standard deviation equals $\sqrt{\text{mean}}$, so this is the expected $\pm k\sigma$ band (default $k = 1$). The flagged fraction is a compact quality metric.

  > **Interpretation.** $\sqrt{\text{mean}}$ equals the true shot-noise σ only for raw photon counts; for count rates (cps) the envelope is a useful heuristic.

- **Running average.** Sliding-window mean and standard deviation (time- or point-based window). The time-based window handles irregular or non-uniform sampling; a wide window

(30–60 s) separates slow drift from shot noise.

- **Block-variance (blocking) analysis.** The standard error of the mean vs. block size: flat for an uncorrelated trace; a rise-then-plateau is the signature of positive correlations (slow modes, as in reline). The full curve is returned with a correlation flag (robust plateau-vs-small-block ratio, default threshold 1.5 — a general heuristic, not a system-specific cutoff).
- **Count-rate histogram.** Fitted with a Gaussian and/or Poisson distribution (Freedman–Diaconis binning, Sturges fallback). The key diagnostic is the **Fano factor** (variance/mean in count space): ≈ 1 is shot-noise limited (ideal Poisson); > 1 signals excess fluctuations (slow modes, dust, drift). Reduced χ^2 of each fit and the coefficient of variation (vs. the shot-noise expectation $1/\sqrt{N}$) are also reported. See the Advanced Guide for the equation.
- **Stationarity test.** An augmented Dickey–Fuller test gives a p-value for whether the trace is stationary; non-stationarity indicates drift or slow-mode behaviour. Requires `statsmodels`.

9.2 SLS I·sin(θ) Diagnostic

For an ideal, isotropic, dust-free scattering volume, I·sin(θ) is flat across angle. Curvature, asymmetry about 90°, or upturns at the extremes reveal alignment problems, stray light, or dust. Curves can be shown **absolute** (comparing concentrations) or **normalized** (each divided by its mean — for comparing solvents, e.g. toluene vs. water, where absolute scale is meaningless).

> **Tip.** A flat I·sin(θ) is much harder to achieve with water than toluene. Overlaying normalized curves for both is an effective alignment/stray-light check.

9.3 Synthetic Data Generator

Generates artificial light-scattering data with **known ground truth** — to validate the analysis (recover a known answer) or produce test files. The forward model uses the **same** q, Stokes–Einstein, optical-constant, Rayleigh, and Zimm physics as the analysis code, so loading and analysing the output recovers what you put in (no convention mismatch).

Inputs. *DLS* and *SLS* need different physical parameters, so they have separate input blocks that share the optical **conditions** (wavelength, n, temperature, viscosity) and the angle/concentration grid:

- **DLS sizes** — a populations table (median Rh, relative scattering weight, log-normal spread; one row = monomodal, two = bimodal, …) plus the coherence factor β, noise level, and point count. The populations *are* the modality — there is no separate "bimodal" option, and the CONTIN/NNLS distribution is something you then recover by analysing the generated correlogram, not a generated artifact.
- **SLS sample** — Mw, Rg, A_2, and dn/dc.
- **Calibration** — *Default* writes calibration metadata into a saved SLS file (and, on workspace injection, prefills the session calibration so Mw is absolute); *Uncalibrated* omits it, so an analysis run is flagged arbitrary-scale.

Artifacts. Tick **Preview** and/or **Save** per artifact:

Artifact	What it tests	Saved as
Correlogram (one angle)	cumulant / single / double / KWW / CONTIN / NNLS	generic two-column CSV
Count-rate trace	trace diagnostics (outliers, drift, Fano, ADF)	two-column CSV
Multi-angle DLS (all angles)	Γ-vs-q^2	multi-angle `.ASC` (+ a trace per angle)
SLS — full Zimm set	Zimm / Berry (thermodynamic)	multi-concentration SLS `.csv`
SLS — single concentration (angular)	Debye / Guinier (apparent)	SLS `.csv`
SLS — single angle (conc. series)	calibration-free A_2 / single-angle Mw	SLS `.csv`
Full Mw-series test set	everything (a homologous series)	a Clean + a Messy folder

Preview draws the *raw generated data* (the analysis figures come from loading the data into the analysis tabs). **Save** writes loadable instrument files into the chosen folder. **Add to workspace as sample** injects a coherent sample (the primary DLS artifact, the primary SLS artifact, and any trace) under a parsed `name / solvent` identity, so DLS + SLS connect for $\rho = R\mathrm{g}/R\mathrm{h}$ without touching disk. Preview never changes any state.

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10. Running an Analysis

This section summarizes, procedurally, how to use the three analysis tabs. For the equations, derivations, and literature behind each method, see the Advanced Guide.

10.1 DLS tab

The DLS module extracts particle dynamics — decay rates, diffusion coefficients, and hydrodynamic radii — from a measured $g_2(\tau)-1$. The tab is organised into **sub-tabs**, each a persistent full-size view (running one never clears another):

- **Correlogram** — parametric fits (cumulant / single / double / KWW). The correlogram is shown in **four scales at once** — a large lin-log view plus lin-lin / log-lin / log-log thumbnails; **double-click a thumbnail to promote it** to the main view. A residual panel sits below, and **Cumulant** and **KWW** fits each populate their own results table. The raw correlogram appears as soon as you select a measurement, before any fit. The **cumulant order** is set here (seeded from Settings; the per-run value wins). The **cumulant method** is a global choice in **Settings → Cumulant method**: *Nonlinear* (Frisken, the default — fits g_2-1 directly with a floating baseline, more robust to drift and noisy data) or *Linear* (Koppel — the classic log-polynomial fit). It applies to every cumulant-based step (cumulant, Γ vs q^2, replicate averaging); the table shows which method produced each result (and its fitted baseline). Switching the method clears existing cumulant-based results — you are asked to confirm first, and distributions, SLS, and any hand-entered Rh are kept. A **delay window** (min/max, in µs) restricts which lags are fitted. Separately, **Settings → Skip initial lag**

channels drops the first N channels (the shortest lags) from *every* DLS fit — set it to your correlator's afterpulsing count when a fit is dragged to a falsely small size by a noisy first channel (default 0; the skip and the delay-window minimum compose, the later of the two winning).

- **Distribution** — NNLS / CONTIN / Lognormal (tick one or more **methods** to compute, with the Rh/Γ axis toggle). These recover a full distribution of decay rates without assuming a functional form; NNLS resolves well-separated peaks but is spiky, CONTIN is smoother and generally better for broad/real data, and Lognormal is a robust single-mode default for a known-unimodal sample. Resolved **peaks are labelled** on the plot, and a **residuals panel** (data − reconstructed g_2-1 vs delay time) sits below it. The **Rh grid** (min / max / points) and the **CONTIN L-curve** α range are set here (seeded from Settings; the per-run value wins). Resolved peak values are not written out as prose — the Distribution sub-tab has its own **Peak results** table (rows *1st peak / 2nd peak / …*, one colour-matched column per ticked measurement·method, cells reading e.g. *"12.3 nm · 87%"* — the peak *R*h and its intensity-weighted area), mirroring the Correlogram results panel; the same values also roll up into the workspace-wide **Summary** sub-tab.
- **Γ vs q²** and **D vs c** — sample-level analyses. "Γ vs q^2 (sample, multi-angle)" tests whether motion is genuinely diffusive and returns D (and *R*h); "D vs c (sample, multi-conc.)" extrapolates the apparent diffusion coefficient to zero concentration, returning the infinite-dilution *R*h,0 and the interaction parameter *k*D. Both operate on the selected sample's DLS measurements (of the current Mw fraction). Each tab carries a **"Measurements (tick to include in the fit)"** points table — one row per measurement (Angle · c · Γ · *D*app for Γ vs q^2; c · Angle · *D*app for D vs c) — and the **tick boxes choose which points enter the fit** (replacing the older click-on-the-plot exclusion). The fit **refits live** as you tick, unticked points grey out on the plot, and **Run** stays disabled until at least two distinct angles / concentrations are ticked (a status line reads *"Tick at least two distinct angles to run."* instead of raising an error dialog). Scalar outputs fill a **Results** table (D, *R*h, R^2, *Diffusive?* for Γ vs q^2; D_0, *R*h,0, *k*D, R^2 for D vs c). The Γ and D used here come from an **internal 2nd-order cumulant fit** of each correlogram — independent of any saved Correlogram/Distribution result — using the global skip-channels and cumulant method from Settings.
- **Depolarized (DDLS)** — pairs a VV with a VH correlogram at each angle (tag each correlogram's polarization in the Data tab) to separate translational from rotational diffusion, and fits sphere and rigid-rod shape models for particle dimensions. Run with *Run DDLS*; the sphere model needs the solvent viscosity. The rotational-diffusion results and the sphere/rod shape-model outputs each fill their own **results table**.
- **Summary** — a workspace-wide, exportable results table that **persists across save/reload**. A *per-measurement* table (one row per measurement) shows cumulant Rh + PDI, single/KWW Rh, and NNLS/CONTIN peaks as `Rh (Int %)` (Int % is the intensity-weighted peak area, *not* a mass/weight percent). A *sample-level Rh* table collects replicate averages, Γ–q^2 (q→0, **apparent**) and D–c (c→0, **thermodynamic**) — the **Rh Type** column marks that distinction and **From** lists the contributing measurements. The left checklist is shared with the Correlogram/Distribution tabs and doubles as a **"Ticked only"** filter; **Export CSV...** writes one long/tidy row per result or peak. Rows fill in as you run fits and averages (e.g. *Average derived results* from the sidebar).

Co-plotting several measurements. Both the Correlogram and Distribution tabs carry a **"Measurements to plot"** checklist of every loaded DLS measurement (any sample, angle, concentration, or Mw fraction); tick several to overlay them on one axes for comparison. The measurement selected in the sidebar is ticked automatically, and the selection is **shared between the two tabs**. **Run** fits every ticked measurement; on the Distribution tab every ticked **method** is computed for every ticked measurement. Plotting many heavy fits (especially CONTIN) at once is slow, so tick only what you want to compare.

Every single-axes DLS/SLS analysis plot carries an **axis-control bar** beneath it: per-axis linear/log toggles, explicit min/max boxes, and an autoscale button.

> **Delay-time window + baseline region.** A single shared analysis region (set on the **Correlogram** sub-tab and used by all per-measurement DLS methods, on every ticked measurement) defines the fit window and the baseline region. Set them by typing values or by **dragging the markers on the correlogram**: the green dotted lines are the window (points outside are ignored — useful to trim a noisy long-delay tail), and the grey dashed/shaded span is the **baseline region** used by the distribution methods (parametric fits don't use a baseline). When a window and a baseline marker overlap, tell them apart by their **offset caret handles**: the **window** handles are **green carets along the top** of the plot, the **baseline** handles are **grey carets along the bottom** — grab the top or bottom half of the plot to pick the one you want. Defaults: full range (the union of all ticked measurements), and the last 25 % of delays for the baseline. A ticked measurement whose lag axis doesn't overlap the window (e.g. a millisecond-lag set co-plotted with a microsecond-lag one) can have too few points inside — it is then **skipped with a note**; drag the markers to a suitable range.

Procedure. Select a measurement (or tick several), set the delay window and baseline, choose a method (or tick the distribution methods you want), click **Run**, and read the result in the table and plot. See the Advanced Guide for the equations and theory.

10.2 SLS tab

The SLS module extracts the weight-average molecular weight $M\mathrm{w}$, the radius of gyration $R\mathrm{g}$, and the second virial coefficient A_2. It first converts raw intensities to absolute excess Rayleigh ratios (via the calibration panel), then constructs the Zimm, Berry, or Debye plots.

Calibration panel. The program computes its own calibration constant $k\mathrm{c}$ from a single calibrant measurement — the intensity of a standard liquid (usually toluene), the angle it was measured at, and that standard's geometry-aware Rayleigh factor. The constant is shown and can be edited. Any value stored in the instrument file is kept only for reference and **never** enters the calculation. If no $k\mathrm{c}$ can be supplied, the analysis still runs on an arbitrary scale, flagged uncalibrated — then $M\mathrm{w}$ and the absolute A_2 are marked unreliable, but $R\mathrm{g}$ and the calibration-free product $2A_2M\mathrm{w}$ remain valid.

Methods available.

- **Single-concentration (apparent) Debye** — at one concentration, gives an apparent $M\mathrm{w}$ and apparent $R\mathrm{g}$. A single angle yields only an apparent $M\mathrm{w}$ and no $R\mathrm{g}$. Apparent values are never presented as final.
- **Zimm / Berry double extrapolation** — combines several concentrations and angles and extrapolates to both zero angle and zero concentration, giving $M\mathrm{w}$, $R\mathrm{g}$, and A_2 at once. The result is marked **thermodynamic**. Berry linearizes the high-$qR\mathrm{g}$ regime (large $M\mathrm{w}$) better than Zimm; the program reports which construction it used.

- **Calibration-free $\mathbf{A}_2$** — obtains the product $2A_2M$w from ratios of excess intensities across concentration at a fixed angle, without any absolute calibration. Valuable when absolute calibration is uncertain.
- **Guinier** — a single-concentration route to Rg from the log intensity, scale-independent and reliable even on an uncalibrated run.
- **Rayleigh series** — the excess Rayleigh ratios themselves.

  Spacing constant (Zimm/Berry grid). Set the purely **aesthetic** spacing constant in the **Spacing k** box (or press **Suggest** for the automatic value). It only changes how spread-out the curves look — Mw, Rg, and A_2 come from the real fit and are unaffected.

Data masking. You can hide whole angles/concentrations or individual points (click to mask, greyed and persisted); an over-masked Zimm is redirected to Debye/Guinier. A manual-Mw override is available and is never overwritten by a re-analysis.

Depolarization correction. For optically anisotropic scatterers, measuring the depolarized (VH) component enables a Cabannes correction that keeps Mw from being overestimated, and yields the optical anisotropy as a result in its own right. Enter IVH and press **Compute depolarization**; the correction is **display-only** and is never written back to the sample's Mw.

Results. Each method fills a two-column **Results** table — Zimm/Berry give Mw, Rg, A_2 and R^2; Debye/Guinier the apparent values; single-angle an apparent Mw; calibration-free A_2 the $2A_2M$w product. The depolarization correction has its **own table** (ρv, ρu, δ^2, Cabannes f, anisotropic %). The Rayleigh series has no scalar table — it shows a one-line status (*"Excess Rayleigh ratio for N concentrations."*) alongside its plot. Switching method or measurement **keeps the last result on screen** rather than blanking it, so you can compare as you go.

Procedure. Set the calibration in the panel (or accept the uncalibrated fallback), choose the fraction (if any), select a method, mask any bad points/angles, click **Run analysis**, and read Mw / Rg / A_2 with their apparent/thermodynamic and calibrated/uncalibrated flags. See the Advanced Guide for the equations and theory.

10.3 Cross-Sample tab

The Cross-Sample module computes quantities that link samples or modules. It is *aggregate-scoped*: instead of the shell navigator, it shows an **include/exclude** list of every sample that has both DLS and SLS (all start included).

$\rho = R\mathrm{g}/R\mathrm{h}$. The shape parameter links Rg (from SLS) and Rh (from DLS) of the same sample. Each included sample (or *fraction*) gets one row: Rg, Rh, ρ, a **Shape** label naming the likely architecture, and whether ρ is **apparent** or **thermodynamic**. The program checks that the two measurements share polymer, solvent, and temperature.

Because a sample can have several Rg and Rh values, you **choose the source** for each (a dropdown per row), and the program never picks silently:

- ***R*g** (SLS): a Zimm or Berry double extrapolation (*thermodynamic*), or a single-concentration Debye/Guinier (*apparent*).
- ***R*h** (DLS): a $c\to0$ D-vs-c extrapolation (*thermodynamic*), a $q\to0$ Γ-vs-q^2 extrapolation at one concentration (*apparent*), a single-condition cumulant (*apparent*), or a **CONTIN/NNLS distribution peak** (*apparent*).

- A **hand-entered** value is always available for either, and is never overwritten by a later re-analysis.

Picking a peak. If you have run a CONTIN or NNLS distribution for a measurement in the DLS tab, each resolved population appears as its own Rh source in the dropdown, so you choose which population feeds ρ. The dominant peak can be the default; a minor peak is selectable but never chosen automatically. The percentages are intensity fractions, not number or mass fractions. Distribution peaks are surfaced only from a distribution you have already computed.

The **default** source is the most-extrapolated the data support (a labelled choice), with fit quality as the tiebreak. **ρ is flagged apparent if *either* input is apparent.** For a physically meaningful shape parameter, prefer the thermodynamic (infinite-dilution) source on both axes.

Scaling laws (Rg–Mw, $\mathbf{A_2}$–Mw). The **Scaling** sub-tab shows two stacked plots over the included samples (one point per sample/fraction), using each sample's effective Mw (a hand-entered Mw wins) chosen in the same source panel. The lower plot is always A_2–Mw; the upper plot is a **size** scaling whose axis you swap between Rg–Mw (SLS) and Rh–Mw (DLS) with the **Show Rh–Mw / Show Rg–Mw** button. Below each plot the tab prints a plain-language reading of the fitted exponent. Two cautions the tab surfaces: an **uncalibrated** Mw is on an arbitrary scale (its horizontal position is meaningless — enter a trusted Mw instead), and A_2 can be **negative** (below the θ-point), in which case that point cannot sit on the log A_2 axis and is excluded. Curating which samples form the series is the user's responsibility — mixing polymers or solvents on one fit is not meaningful.

Procedure. Include the samples you want, choose the Rg / Rh / Mw source per row, read ρ and its Shape label in the table, and switch to the Scaling sub-tab for the log–log plots and fitted exponents. See the Advanced Guide for the equations and theory.

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11. Exporting Results

Every analysis result can be written to a CSV that OriginLab imports directly. Each file carries the four Origin label rows — **Long Name, Units, Comments, Parameters** — so axis labels and units populate on import. Tabular outputs (correlogram fits, distributions, Rayleigh ratios) are written column by column with units; scalar results (Mw, Rg, Γ, …) as single-value labelled columns. Zimm datasets use a wide layout (one $Kc/\Delta R\theta$ column per concentration, the concentration in the Parameters row). Missing values are empty cells. An uncalibrated result writes "uncalibrated, arbitrary scale" into the **Comments** cell of the affected columns only — never changing the row count, so the Origin import stays intact.

From the interface. Each analysis view has an **Export CSV…** button that writes the result currently on screen: the DLS **Correlogram**, **Distribution**, Γ **vs q**^{2}, **D vs c**, and **Depolarized (DDLS)** sub-tabs (the DDLS file carries the per-angle ΓVV/ΓVH and Dr, the combined Dt/Dr/Rh,t/τrot, and — when available — the rod and sphere shape dimensions, each flagged "MODEL … not measured" in its Comments cell); the SLS **Analysis** panel (the button exports the last method you ran — Zimm/Berry, Debye, Guinier, single-angle, calibration-free A_2, or the excess Rayleigh series); and the Cross-Sample **Scaling** tab (the Rg/Rh–Mw and A_2–Mw points and fits). The button is enabled once a result has been computed. When a view holds several curves at once (an NNLS+CONTIN distribution overlay, or both size and A_2 scaling plots), the export writes one file per curve, suffixed by method/quantity.

12. Visualization

Every analysis can be plotted from the same result object it returns, so what is plotted is exactly what was computed: correlogram fits with residuals, size distributions, Γ-vs-q^2 diagnostics, Debye plots, and the full Zimm/Berry grid. Distributions from different methods can be overlaid (e.g. NNLS and CONTIN on one axis), making their trade-off visible at a glance.

Data-quality and reliability **flags are interface overlays**, not part of the figure: a saved plot image is always clean. Where a result is apparent or uncalibrated, the interface marks it next to the plot, so a number can't be read off a chart without its caveat.

Appearance (Settings tab). The **Theme** is *System* (follows your OS light/dark setting) by default; choosing *Light* or *Dark* explicitly **overrides** the OS so the application stays in that mode regardless of system preferences. The **Plot palette** (Okabe–Ito colourblind-safe / tab10 / grayscale) applies to plots drawn afterwards. Within the analysis tabs, the **stacked plots and their control columns are resizable** — drag the **splitter grips** (the visible three-dot handles) between them, including the fit/residual divider, which carries a persistent grip line. The workspace panel on the left is **resizable** (drag the divider) and long sample names show in full on hover; the window itself can be freely resized (tab contents scroll when the window is small).